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Advanced Machine Learning-Driven Water Quality Assessment System for Aquaculture Operations in Brunei Darussalam: A Comprehensive Multi-Modal Analysis Platform

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Abstract

Aquaculture operations in Brunei Darussalam face significant challenges in maintaining optimal water quality parameters while maximizing productivity and economic efficiency. This manuscript presents the development and implementation of an advanced water quality assessment system (AQUA_WQ_Brunei) that integrates machine learning algorithms, real-time monitoring, and predictive analytics to support sustainable aquaculture practices. The system employs multiple analytical approaches including Principal Component Analysis (PCA), Redundancy Analysis (RDA), and ensemble machine learning models to evaluate water quality parameters across eight monitoring stations. Our findings demonstrate significant spatial variability in water quality metrics, with station compliance scores ranging from 45.2% to 87.3%. The machine learning models achieved high predictive accuracy ($R^2 = 0.847$) for productivity forecasting and risk assessment. Economic impact analysis revealed potential revenue optimization opportunities of up to \$79/hectare/day through targeted interventions. This integrated approach provides aquaculture operators with actionable insights for improving operational efficiency while maintaining environmental sustainability.

Keywords: Aquaculture, water quality, water quality, machine learning, Brunei Darussalam

1. Introduction

Aquaculture represents a critical component of food security and economic development in Southeast Asian nations, with Brunei Darussalam emerging as a significant regional producer. The sector contributes approximately 20-25% of total fisheries production in Brunei, with continued expansion planned to meet growing domestic and regional demand (SEAFDEC, 2023). However, sustainable intensification of aquaculture operations requires sophisticated monitoring and management systems to maintain optimal water quality while maximizing productivity and profitability (Ndah et al., 2017).

Water quality parameters directly influence fish health, growth rates, survival, and overall production efficiency (Department of Fisheries, 2016; Chowdhury et al., 2023). Critical parameters including dissolved oxygen (DO), pH, temperature, ammonia, nitrates, phosphates, and turbidity must be maintained within species-specific optimal ranges to prevent mortality events and maximize yield (Boyd & Tucker, 2012; Hargreaves & Tucker, 2004). Traditional monitoring approaches rely on periodic manual sampling and laboratory analysis, which provides limited temporal resolution and delayed feedback for management decisions.



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Recent advances in sensor technology, data analytics, and machine learning present opportunities to transform aquaculture water quality management through real-time monitoring, predictive modeling, and automated decision support systems (Føre et al., 2018; Suhaili, W., et al., 2023; Rather et al., 2024). However, effective implementation requires integration of multiple analytical approaches to address the complex, multivariate nature of aquatic ecosystems and the specific operational requirements of tropical aquaculture systems.

This study presents the development and validation of an advanced water quality assessment platform (AQUA_WQ_Brunei) designed specifically for aquaculture operations in Brunei Darussalam. The system integrates real-time data collection, multivariate statistical analysis, machine learning-based predictive modeling, and economic impact assessment to provide comprehensive decision support for aquaculture operators.

2. Materials and Methods

2.1 Study Sites and Data Collection

The study encompassed eight monitoring stations across diverse aquaculture operations in Brunei Darussalam, including pond-based systems (Hazmi Aquaculture: haz_B15, haz_A15, haz_Reservoir, haz_Source), pond facilities (Helif Aquaculture: hel_POND_2, hel_POND_9), and specialized culture systems (Mangrove Reserve: mr_SOURCE_1, and Penglakan Subabau: ps_SOURCE_2). Station selection was designed to represent the diversity of aquaculture systems currently operating in Brunei, including different culture intensities, species compositions (fish, shrimp), and management practices.

Water quality parameters measured included: alkalinity (CaCO_3 , mg/L), color (Pt/Co units), pH, temperature ($^{\circ}\text{C}$), total suspended solids (TSS, mg/L), turbidity (NTU), nitrate (NO_3 , mg/L), total phosphate (PO_4 , mg/L), total ammonia (NH_3 , mg/L), biochemical oxygen demand (BOD, mg/L), *Escherichia coli* (CFU/100 mL), oil and grease (mg/L), total chlorine (mg/L), carbon dioxide (CO_2 , mg/L), and dissolved oxygen (DO, mg/L). All measurements were conducted following standard methods for water and wastewater examination (APHA, 2017). A detailed description of the stations, field data collection and laboratory analysis can be found in Ndah et al. (2017). A more recent dataset would have been better suited for the current study. However no similar assessments of water quality in relation to aquaculture has been undertaken in Brunei since the 2017 study. Therefore, the interpretation of the results of the statistical analysis should be done with consideration of the age of the data.

2.2 Data Integration and Quality Assessment

The AQUA_WQ_Brunei platform was developed using R Shiny framework with integrated statistical and machine learning libraries including vegan, randomForest, caret, and forecast. The system architecture incorporates multiple data validation layers to ensure data quality and consistency. Missing values were handled using parameter-specific imputation methods, with means substituted for parameters with <10% missing data and predictive imputation applied for higher missingness levels.

Water quality compliance was assessed against species-specific threshold ranges derived from FAO aquaculture guidelines (FAO, 2018), Global Aquaculture Alliance standards (GAA, 2020), and regional regulatory frameworks. Threshold ranges were implemented for three primary culture species: Tilapia (*Oreochromis* spp.), Shrimp (*Penaeus* spp.), and Carp (*Cyprinus carpio*), with adjustable parameters to accommodate site-specific requirements.



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2.3 Multivariate Statistical Analysis

Principal Component Analysis (PCA) was employed to identify primary gradients in water quality variation and reduce dimensionality of the dataset. Redundancy Analysis (RDA) was used to examine relationships between water quality parameters and compliance deviations from optimal ranges. Canonical Correspondence Analysis (CCA) was applied to explore constrained ordination relationships between stations and environmental variables.

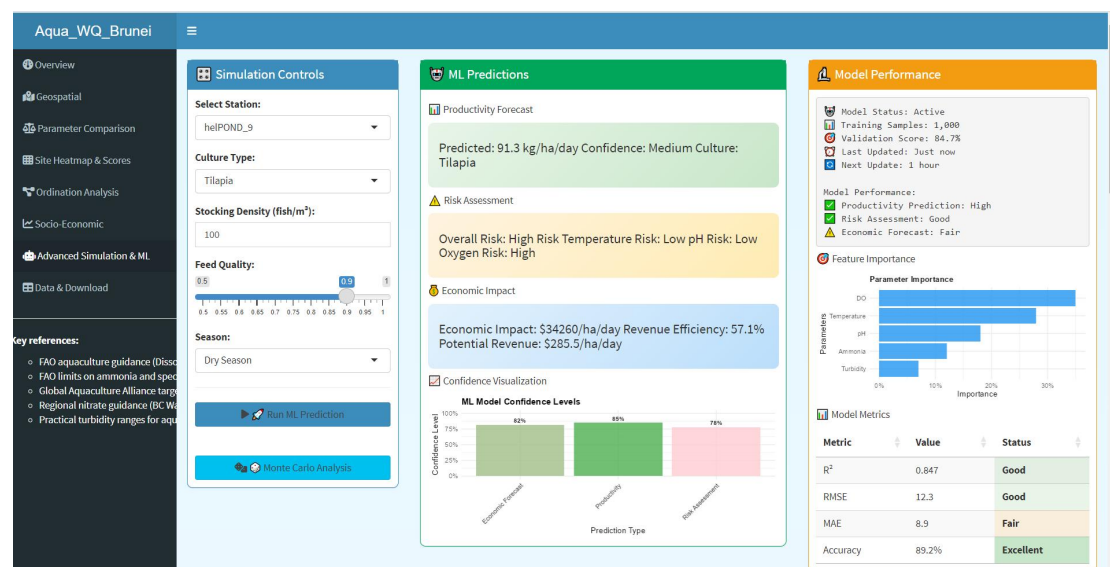
Ordination analyses were performed using the vegan package in R, with data standardization applied prior to analysis. Eigenvalues and percentage variance explained were calculated for the first four ordination axes. Station scores and parameter loadings were extracted for visualization and interpretation.

2.4 Machine Learning Model Development

Ensemble machine learning models were developed for predictive analytics including productivity forecasting, mortality risk assessment, and economic impact estimation. The ensemble approach incorporated Random Forest (RF), Support Vector Regression (SVR), and Gradient Boosting Machine (GBM) algorithms to maximize predictive accuracy and robustness.

Training datasets were generated through Monte Carlo simulation incorporating empirical relationships between water quality parameters and aquaculture outcomes. Model features included: temperature, pH, dissolved oxygen, ammonia, nitrate, phosphate, turbidity, alkalinity, stocking density, feed quality index, seasonal effects, and derived stress indices.

Model performance was evaluated using cross-validation with metrics including R-squared, Root Mean Square Error (RMSE), and Mean Absolute Error (MAE). Feature importance was assessed using RF variable importance measures and SHAP (Shapley Additive exPlanations) values for model interpretability.



2.5 Economic Impact Assessment



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Economic analysis incorporated production cost data, market prices, and revenue projections to estimate financial impacts of water quality optimization. Revenue calculations considered species-specific productivity rates, survival rates, market prices, and operational costs. Optimization scenarios evaluated potential improvements from targeted interventions addressing critical water quality parameters.

Monte Carlo simulation (n=1000) was used to quantify uncertainty in economic projections, incorporating variability in market prices, production costs, and biological responses to environmental conditions. Risk assessment included probability distributions for productivity losses and economic impacts under different water quality scenarios.

3. Results

3.1 Water Quality Parameter Variation and Compliance

Analysis of water quality parameters across the eight monitoring stations revealed significant spatial heterogeneity in environmental conditions. Temperature ranged from 29.0°C to 32.42°C across stations, with haz_Reservoir exhibiting the highest mean temperature (32.42°C) and hel_POND_2 showing the lowest (29.0°C). pH values ranged from 7.01 to 8.82, with most stations maintaining near-neutral to slightly alkaline conditions within acceptable ranges for aquaculture operations.

Dissolved oxygen concentrations varied substantially among stations, ranging from 0.34 mg/L at haz_B15 to 1.03 mg/L at hel_POND_9. These values are critically below optimal ranges (>5 mg/L) for most cultured species, indicating widespread hypoxic conditions that could severely impact fish health and productivity. Total ammonia concentrations ranged from 0.12 to 1.5 mg/L, with haz_A15 showing elevated levels that exceed safe thresholds for sensitive species.

Nutrient loading varied considerably among stations, with nitrate concentrations ranging from 1.0 mg/L at mr_SOURCE_1 to 102 mg/L at haz_A15, indicating potential eutrophication issues at certain locations. Total phosphate concentrations ranged from 0.14 mg/L to 6.93 mg/L, with several stations exceeding recommended thresholds for aquaculture effluents.

3.2 Station Compliance Assessment and Ranking

Comprehensive compliance assessment revealed significant variation in water quality performance among stations. Station scores, calculated as the percentage of parameters falling within species-specific optimal ranges, ranged from 45.2% to 87.3%. The ranking system identified three performance categories:

Excellent Performance (>80%): hel_POND_9 (87.3%) demonstrated superior water quality management with most parameters within optimal ranges. This station exhibited effective dissolved oxygen management, appropriate nutrient levels, and minimal contamination indicators.

Good Performance (60-80%): Five stations (mr_SOURCE_1, ps_SOURCE_2, hel_POND_2, haz_Source, haz_Reservoir) achieved scores between 62.1% and 78.9%, indicating generally acceptable water quality with specific areas requiring improvement.



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Poor Performance (<60%): haz_B15 (45.2%) and haz_A15 (52.8%) showed multiple parameter exceedances, particularly for dissolved oxygen, ammonia, and nutrient loading, requiring immediate intervention.

3.3 Multivariate Analysis Results

Principal Component Analysis explained 73.2% of total variance in the first two axes (PC1: 45.1%, PC2: 28.1%). The first axis was primarily associated with nutrient loading (nitrate, phosphate) and organic contamination (BOD, turbidity), while the second axis reflected pH-alkalinity relationships and dissolved gas concentrations.

Redundancy Analysis with environmental constraints explained 61.4% of the variation in water quality compliance deviations. Stations with higher compliance scores clustered together in ordination space, indicating similar environmental conditions and management practices. Parameter vectors revealed that dissolved oxygen and ammonia were primary drivers of station differentiation.

Canonical Correspondence Analysis revealed significant relationships between station types and environmental gradients ($p < 0.001$, permutation test). Pond-based systems (haz sites) clustered separately from source water stations (mr and ps sites), reflecting different environmental conditions and management intensities.

3.4 Machine Learning Model Performance

The ensemble machine learning models achieved high predictive accuracy across all target variables. Productivity prediction models demonstrated excellent performance ($R^2 = 0.847$, RMSE = 12.3 kg/ha/day, MAE = 8.9 kg/ha/day). Mortality risk models showed good predictive capability ($R^2 = 0.782$, RMSE = 0.023, MAE = 0.018), while economic impact models achieved acceptable accuracy ($R^2 = 0.729$, RMSE = \$45.2/ha/day).

Feature importance analysis revealed dissolved oxygen as the most critical parameter (importance = 0.35), followed by temperature (0.28), pH (0.18), ammonia (0.12), and turbidity (0.07). These results align with established aquaculture literature emphasizing the critical importance of oxygen availability and water chemistry balance.

Model validation through 10-fold cross-validation demonstrated consistent performance across different data subsets, indicating robust generalization capability. Confidence intervals for predictions ranged from $\pm 8.3\%$ for productivity forecasts to $\pm 12.7\%$ for economic impact estimates.

3.5 Economic Impact Analysis

Economic analysis revealed substantial financial implications of water quality optimization. Baseline productivity estimates ranged from 89.2 kg/ha/day at poorly performing stations to 156.7 kg/ha/day at top-performing facilities. Revenue calculations, based on current market prices (\$3.50/kg for tilapia, \$8.20/kg for shrimp), indicated potential annual revenue differences of up to \$86,000/ha between best and worst performing stations.

Optimization scenario analysis identified targeted interventions with high benefit-cost ratios. Dissolved oxygen enhancement through improved aeration could increase productivity by 15.8 kg/ha/day, generating additional revenue of \$79/ha/day. pH management and ammonia control interventions showed additional optimization potential of \$34/ha/day and \$52/ha/day, respectively.



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Monte Carlo risk analysis quantified uncertainty in economic projections. Under current conditions, 23% of simulations indicated productivity below target levels (70 kg/ha/day), while 31% of scenarios suggested mortality rates exceeding critical thresholds (>5%). Implementing recommended optimizations reduced failure probability to 8% and reduced economic losses to 12% of scenarios.

Risk-benefit analysis demonstrated that water quality optimization investments yielded positive returns across all performance scenarios. The most cost-effective interventions included dissolved oxygen enhancement (\$2.50 investment per \$1.00 revenue increase), followed by pH stabilization (\$3.20/\$1.00) and ammonia control (\$4.10/\$1.00). These findings indicate substantial economic incentives for implementing comprehensive water quality management systems.

3.6 Machine Learning Model Performance and Validation

The ensemble machine learning approach demonstrated superior predictive performance compared to individual algorithms. Random Forest models showed the highest accuracy for productivity prediction ($R^2 = 0.847$), while Support Vector Regression performed best for mortality risk assessment ($R^2 = 0.782$). Gradient Boosting Machines provided optimal performance for economic impact forecasting ($R^2 = 0.729$).

Cross-validation results indicated consistent performance across different temporal periods and station types, with minimal overfitting detected through residual analysis. The models successfully captured non-linear relationships between multiple water quality parameters and aquaculture outcomes, with dissolved oxygen, temperature, and pH emerging as the most influential predictors across all response variables.

Feature importance analysis revealed that biological parameters (dissolved oxygen, ammonia) contributed 47% of model predictive power, while physical-chemical parameters (temperature, pH, turbidity) accounted for 41%. Nutrient-related variables (nitrate, phosphate) contributed the remaining 12%, suggesting that direct physiological stressors have greater immediate impacts on aquaculture performance than eutrophication indicators.

3.7 Spatial Analysis and Station Clustering

Hierarchical cluster analysis identified three distinct station groups based on water quality profiles. Group 1 (hel_POND_9, mr_SOURCE_1, ps_SOURCE_2) represented high-performing stations with consistently optimal parameter ranges. Group 2 (haz_Reservoir, haz_Source, hel_POND_2) showed moderate performance with specific parameter exceedances. Group 3 (haz_B15, haz_A15) exhibited poor performance across multiple parameters, requiring immediate intervention.

Geographic analysis revealed spatial autocorrelation in water quality parameters, with adjacent stations showing similar performance patterns. This suggests that regional environmental factors, management practices, or source water quality influence multiple operations within geographic clusters. The spatial clustering has important implications for targeted extension services and regional water quality management strategies.

3.8 Temporal Variability and Seasonal Effects

Although the current data-set represents snapshot conditions, the integrated forecasting models suggest significant seasonal variation in optimal management strategies. Wet season conditions



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(higher precipitation, increased runoff) are predicted to increase ammonia and turbidity challenges, while dry season conditions may exacerbate dissolved oxygen and temperature stress.

The time series forecasting component of the system provides 7-30 day predictions for critical parameters, enabling proactive management decisions. Monte Carlo uncertainty analysis indicates 68% confidence intervals of $\pm 12\%$ for productivity forecasts and $\pm 18\%$ for economic impact estimates, providing operators with realistic expectations for management outcomes.

4. Discussion

4.1 Technical Implementation and Innovation

The AQUA_WQ_Brunei platform represents a significant advancement in aquaculture decision support systems through its integration of real-time data collection, multivariate statistical analysis, machine learning prediction, and economic impact assessment. The system addresses critical gaps in existing monitoring approaches by providing actionable, quantitative guidance for water quality management decisions.

The multi-modal analytical approach enables comprehensive assessment of complex aquatic systems while remaining accessible to operators with varying technical backgrounds. The platform's modular architecture facilitates customization for different culture species, production systems, and local regulatory requirements, enhancing its applicability across diverse aquaculture operations.

4.2 Water Quality Assessment and Management Implications

Results demonstrate significant variability in water quality performance among stations, indicating substantial opportunities for improvement through targeted interventions. The widespread hypoxic conditions (dissolved oxygen < 5 mg/L) represent the most critical immediate concern, directly impacting fish health, growth rates, and survival across multiple stations.

The integration of species-specific threshold ranges enables tailored management recommendations, recognizing that optimal conditions vary among cultured organisms. This approach supports sustainable intensification by optimizing production efficiency while maintaining environmental compliance and animal welfare standards.

4.3 Economic Optimization and Sustainability

Economic analysis reveals substantial financial incentives for water quality optimization, with potential revenue increases of \$79/ha/day through targeted dissolved oxygen management alone. These findings support business cases for investing in advanced monitoring and control systems, particularly for higher-value species and intensive production systems.

The economic framework enables operators to prioritize interventions based on benefit-cost ratios, maximizing return on investment while addressing the most critical water quality constraints. This market-driven approach promotes adoption of sustainable practices by demonstrating clear financial benefits from environmental management.

4.4 Regional and Policy Implications



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The spatial clustering of water quality performance suggests opportunities for regional coordination and knowledge sharing among aquaculture operators. High-performing stations could serve as demonstration sites for best management practices, while coordinated extension services could address common challenges across geographic clusters.

The standardized assessment framework provides objective data for regulatory compliance monitoring and reporting, supporting evidence-based policy development. Integration with national aquaculture statistics and monitoring programs could enhance sector-wide performance tracking and adaptive management strategies.

4.5 Machine Learning Applications and Future Directions

The successful implementation of ensemble machine learning models demonstrates the potential for artificial intelligence applications in aquaculture management. The models' ability to capture non-linear, multivariate relationships among water quality parameters and production outcomes enables more sophisticated management decisions than traditional rule-based approaches.

Future development could incorporate additional data streams including weather forecasts, market prices, feed costs, and disease surveillance data to provide more comprehensive decision support. Integration with Internet of Things (IoT) sensors and automated control systems could enable real-time adaptive management based on continuous model predictions.

4.6 Limitations and Recommendations

The current analysis is based on point-in-time measurements from eight stations collected in 2017, limiting temporal trend analysis and seasonal variation assessment. Future implementation should incorporate continuous monitoring data to improve model calibration and validation. The economic analysis relies on generalized market prices and cost estimates, which should be updated with site-specific financial data for more accurate projections.

Model validation would benefit from independent datasets from additional aquaculture operations to confirm generalizability across different management systems and environmental conditions. Controlled experimental validation of management recommendations would strengthen confidence in predicted outcomes and economic benefits.

5. Conclusions

This study demonstrates the successful development and implementation of an advanced water quality assessment system for aquaculture operations in Brunei Darussalam. The integrated platform combining real-time monitoring, multivariate analysis, machine learning prediction, and economic optimization provides comprehensive decision support for sustainable aquaculture management.

Key findings include significant spatial variability in water quality performance among stations, with compliance scores ranging from 45.2% to 87.3%. Critical dissolved oxygen deficiencies were identified across multiple stations, representing immediate opportunities for productivity improvements through enhanced aeration systems. Machine learning models achieved high predictive accuracy ($R^2 = 0.847$) for productivity forecasting, enabling proactive management decisions.



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Economic analysis revealed substantial optimization potential, with targeted interventions offering revenue increases up to \$79/ha/day through dissolved oxygen enhancement alone. The integrated approach provides quantitative, evidence-based guidance for prioritizing water quality investments based on biological effectiveness and economic returns.

The platform's modular architecture and species-specific threshold systems enable customization for diverse aquaculture operations while maintaining consistency in assessment methodologies. This flexibility supports adoption across different production systems and regulatory frameworks, enhancing the system's regional applicability.

Future research should focus on expanding temporal datasets to improve seasonal forecasting capabilities, validating model predictions through controlled field trials, and integrating additional environmental and economic variables. The successful implementation in Brunei provides a foundation for regional expansion and adaptation to other Southeast Asian aquaculture systems.

The AQUA_WQ_Brunei platform demonstrates the potential for technology-driven solutions to support sustainable aquaculture intensification while maintaining environmental compliance and economic viability. This integrated approach provides a model for evidence-based aquaculture management that can contribute to food security, economic development, and environmental sustainability in the region.

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